Review & Security

Automated MCQ Exam Portal

Code Review	CodeRabbit Review completed; no actionable comments reported	**PASS**
Security Review	Snyk Final scan: Critical 0; High 21; Medium 21; Low 0	**PASS**
PR Verification	GitHub checks passed; no merge conflicts reported	**PASS**

1. Code Review - CodeRabbit

A GitHub pull request was created for automated review. CodeRabbit completed the review and the PR page reported no actionable comments in the recent review. The PR also showed the repository checks passing.

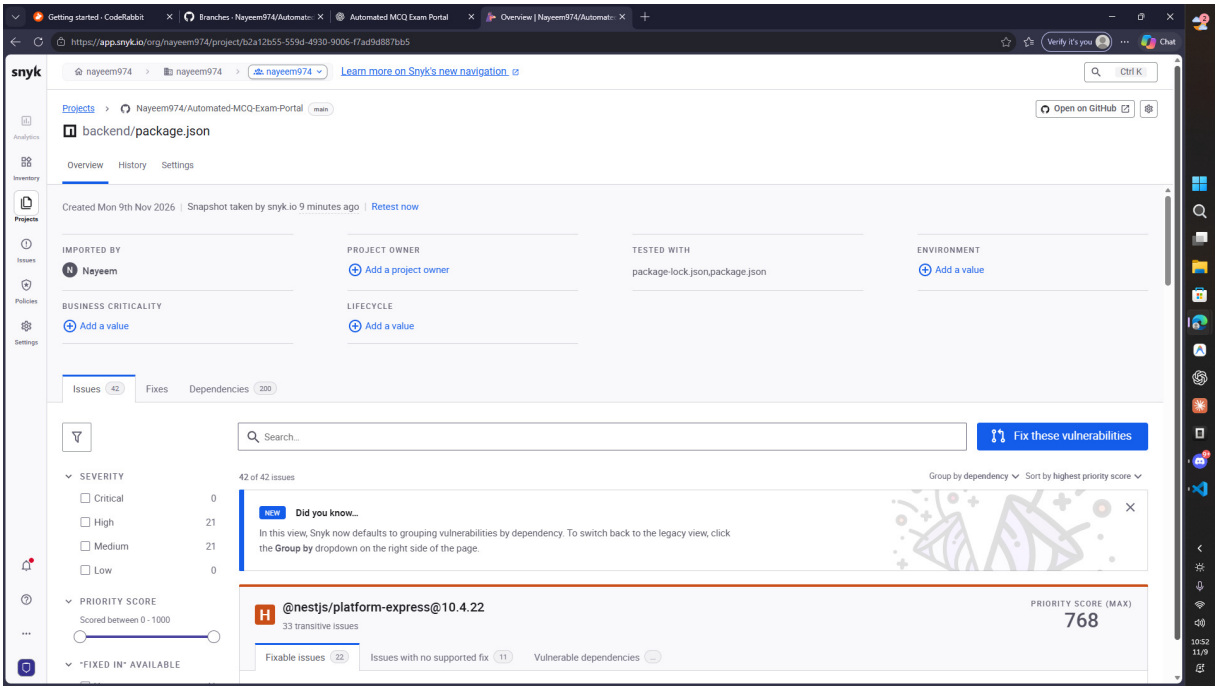
Observed in PR	Result
CodeRabbit review	Completed; no actionable comments
Qodo review	Review output shown on the PR; no issues reported
Backend checks	Passed
Frontend build checks	Passed
Snyk security check	Passed
Vercel deployment check	Passed
Merge status	No conflicts with main

The full GitHub PR screenshot is retained in the project evidence; this report summarizes the verified items so the document stays compact and readable.

2. Security Review - Snyk

Snyk was connected to the GitHub repository and scanned dependency manifests. The initial backend scan identified a Critical dependency vulnerability associated with proxy-addr. A targeted npm override to proxy-addr 2.0.8 was then applied and verified locally.

Final Snyk Result



Final evidence: Snyk project view for backend/package.json on the main branch showing Critical 0, High 21, Medium 21, Low 0.

Severity	Final count
Critical	0
High	21
Medium	21
Low	0

Remediation

The critical dependency issue was addressed with a targeted proxy-addr 2.0.8 override. After the change, the project test suite remained at 46/46 passing tests and the backend build completed successfully. The final Snyk result reduced the Critical count to 0.

Important: the final report does not claim zero vulnerabilities overall because 21 High and 21 Medium findings remain in the Snyk snapshot.

3. Final Review & Security Status

Area	Tool	Result
Code review	CodeRabbit	Completed; no actionable comments
AI-assisted review	Qodo	Review output recorded on the PR
Security scanning	Snyk	Critical 0; High 21; Medium 21; Low 0
Repository verification	GitHub PR checks	All displayed checks passed

Conclusion

The project completed automated code review, AI-assisted review, dependency security scanning, targeted remediation of the identified Critical issue, and final verification through the project test suite and GitHub checks. The final security state is reported exactly as shown by Snyk.